

Nonparametric estimation of causal parameters with binary outcomes

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1 Unconfoundedness

Suppose we have two binary potential outcomes $Y(1), Y(0)$ given by

$$\begin{aligned} Y(1) &= 1\{Y(1)^* \geq 0\} \\ Y(0) &= 1\{Y(0)^* \geq 0\} \end{aligned}$$

where $Y(1)^*, Y(0)^*$ are two potential latent utilities associated with the choice $Y(1) = 1$ if treated and the choice $Y(0) = 1$ if untreated. Without loss of generality, suppose the potential latent utilities are generated as

$$\begin{aligned} Y(1)^* &= m(1, V) - \varepsilon(1) \\ Y(0)^* &= m(0, V) - \varepsilon(0) \end{aligned}$$

where $m(1, V)$ and $m(0, V)$ are nonparametric systematic components of $Y(1)^*$ and $Y(0)^*$ respectively, and the cumulative distribution functions of the error terms $\varepsilon(1), \varepsilon(0)$ are given by

$$\begin{aligned} \mathbb{P}\{\varepsilon(1) \leq z\} &= F_1(z) \\ \mathbb{P}\{\varepsilon(0) \leq z\} &= F_0(z) \end{aligned}$$

We see that the conditional choice probabilities for each potential outcome are given by

$$\begin{aligned} \mathbb{P}\{Y(w) = 1 | V = v\} &= \mathbb{P}\{Y(w)^* \geq 0 | V = v\} \\ &= F_w(m(w, v)) \end{aligned}$$

for $w = 1, 0$.

Suppose that the unconfoundedness assumption holds so that the assignment to treatment $W \in \{1, 0\}$ can be viewed as random, conditional on the observed V .

$$\{Y(1), Y(0)\} \perp W | V \tag{1}$$

Given V , the randomness in $Y(1)$ and $Y(0)$ comes only from $\varepsilon(1)$ and $\varepsilon(0)$ so that the unconfoundedness assumption implies

$$\{\varepsilon(1), \varepsilon(0)\} \perp W | V$$

Under the unconfoundedness assumption, we have

$$\begin{aligned}
P\{Y = 1|V = v, W = w\} &= P\{Y(w) = 1|V = v, W = w\} \\
&= P\{Y(w)^* \geq 0|V = v, W = w\} \\
&= P\{\varepsilon(w) \leq m(w, v)|V = v, W = w\} \\
&= P\{\varepsilon(w) \leq m(w, v)|V = v\} \\
&= F_w(m(w, v))
\end{aligned}$$

for $w = 1, 0$, where we've used the unconfounded assumption on the fourth equality.

Here, the nonparametric systematic component $m(w, v)$ and the cumulative distribution of the error terms F_w are not separately identified. This implies that as long as we have a fully nonparametric specification of $m(w, v)$, we may assume F_1 and F_0 are logistic distributions without loss of generality. To see this, let $Y(1)^*, Y(0)^*$ be generated as

$$\begin{aligned}
Y(1)^* &= \tilde{m}(1, V) - \tilde{\varepsilon}(1) \\
Y(0)^* &= \tilde{m}(0, V) - \tilde{\varepsilon}(0)
\end{aligned}$$

where $\tilde{m}(1, V)$ and $\tilde{m}(0, V)$ are another nonparametric systematic components of $Y(1)^*$ and $Y(0)^*$ respectively, and the error terms both follow logistic distribution

$$\mathbb{P}\{\varepsilon(1) \leq z\} = \mathbb{P}\{\varepsilon(0) \leq z\} = \frac{1}{1 + e^{-z}}$$

Here, if we let $\tilde{m}(w, v) = -\log(F_w(m(w, v))^{-1} - 1)$, the conditional choice probability is given by

$$\begin{aligned}
P\{Y = 1|V = v, W = w\} &= P\{Y(w) = 1|V = v, W = w\} \\
&= P\{Y(w)^* \geq 0|V = v, W = w\} \\
&= P\{\tilde{\varepsilon}(w) \leq \tilde{m}(w, v)|V = v, W = w\} \\
&= P\{\tilde{\varepsilon}(w) \leq \tilde{m}(w, v)|V = v\} \\
&= \frac{1}{1 + e^{-\tilde{m}(w, v)}} \\
&= \frac{1}{1 + e^{\log(F_w(m(w, v))^{-1} - 1)}} \\
&= F_w(m(w, v))
\end{aligned}$$

Thus, the conditional choice probability of the observables can always be written as

$$P\{Y = 1|V = v, W = w\} = \frac{1}{1 + e^{-m(w, v)}}$$

and the conditional choice probability of the potential outcomes can also be written as

$$P\{Y(w) = 1|V = v\} = \frac{1}{1 + e^{-m(w, v)}}$$

for $w = 1, 0$ for some nonparametric function $m(w, v)$. This implies that given the data $\{Y_i, V_i, W_i\}_{i=1}^n$, we may estimate the nonparametric function $m(w, v)$ using the likelihood function

$$\ell(y, v, w) = -y \log(1 + e^{-m(w, v)}) - (1 - y) \log(1 + e^{m(w, v)})$$

and use the estimated $\hat{m}(w, v)$ to obtain causal parameter of interest.

2 Models for the nonparametric systematic part $m(w, v)$

1. We may use a class of simple functions to model $m(w, v)$ with the binary splitting algorithm for estimation. This corresponds to fitting a decision tree classifier with the cross entropy loss function.
2. We may specify $m(w, v)$ as an element of RKHS associated with a given positive definite kernel.
3. We may specify $m(w, v)$ using a neural network. This corresponds to estimating a neural network with a sigmoid output layer and cross-entropy loss function.

3 Estimation of Causal Parameters

Recall that the conditional choice probability for $Y(1)$ and $Y(0)$ are given as

$$P\{Y(w) = 1|V = v\} = \frac{1}{1 + e^{-m(w, v)}}$$

for $w = 0, 1$. The causal parameter of interest is the conditional partial effect defined as

$$\begin{aligned}\tau(v) &= P\{Y(1) = 1|V = v\} - P\{Y(0) = 1|V = v\} \\ &= \frac{1}{1 + e^{-m(1, v)}} - \frac{1}{1 + e^{-m(0, v)}}\end{aligned}$$

if W is binary and

$$\begin{aligned}\tau(w, v) &= \frac{\partial P\{Y(w) = 1|V = v\}}{\partial w} \\ &= \frac{e^{-m(w, v)}}{(1 + e^{-m(w, v)})^2} \frac{\partial m(w, v)}{\partial w}\end{aligned}$$

if W is continuous.¹ For binary W , the estimation of conditional partial effect is straightforward

$$\hat{\tau}(v) = \frac{1}{1 + e^{-\hat{m}(1, v)}} - \frac{1}{1 + e^{-\hat{m}(0, v)}}$$

For continuous W , the conditional partial effect may be estimated by

$$\hat{\tau}(w, v) = \frac{e^{-\hat{m}(w, v)}}{(1 + e^{-\hat{m}(w, v)})^2} \frac{\partial \hat{m}(w, v)}{\partial w}$$

as long as $\frac{\partial \hat{m}(w, v)}{\partial w}$ exists. If $\hat{m}(w, v)$ is estimated using either RKHS or neural network, $\frac{\partial \hat{m}(w, v)}{\partial w}$ exists. If $\hat{m}(w, v)$ is estimated using random forest, $\frac{\partial \hat{m}(w, v)}{\partial w}$ does not exist. One potential remedy for this is to use a structured random forest. In other words, if we model $m(w, v)$ as having either a partially linear or functional-coefficient form, we have

$$m(w, v) = \beta'w + f(v)$$

¹Assuming that $Y(W)$ exists.

or

$$m(w, v) = \beta(v)'w$$

so that the partial derivative $\frac{\partial \hat{m}(w, v)}{\partial w}$ always exists as either $\hat{\beta}$ or $\hat{\beta}(v)$. After estimating the conditional partial effect, we may estimate the average partial effect as

$$\hat{\mathbb{E}}[\hat{\tau}(V)] = \frac{1}{n} \sum_{i=1}^n \hat{\tau}(V_i), \quad \text{or} \quad \hat{\mathbb{E}}[\hat{\tau}(W, V)] = \frac{1}{n} \sum_{i=1}^n \hat{\tau}(W_i, V_i)$$

and the average partial effect on the treated as

$$\hat{\mathbb{E}}[\hat{\tau}(V)|W = 1] = \frac{1}{|I_1|} \sum_{i \in I_1} \hat{\tau}(V_i), \quad \text{or} \quad \hat{\mathbb{E}}[\hat{\tau}(W, V)|W = 1] = \frac{1}{|I_1|} \sum_{i \in I_1} \hat{\tau}(1, V_i)$$

where I_1 is the index set for the treated observations.

4 Considerations

1. Here, we use logistic distribution for the error terms without loss of generality because given the conditional choice probability

$$P\{Y = 1|V = v, W = w\} = F_w(m(w, v))$$

$m(w, v)$ and F_w are not separately identified. If the causal parameter of interest is the conditional partial effect or average partial effect, there is no need to separately identify $m(w, v)$ and F_w .

2. If we add structure to the nonparametric systematic part $m(w, v)$, then assuming logistic distribution for the error term comes at a loss of generality. However, this might be beneficial for interpretation as well as for computation of partial effects for continuous treatments.